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BRIEF REPORT

Quadrivalent HPV Vaccine Effectiveness Against Cervical Intraepithelial Lesion Grade 2 or Worse in Norway: A Registry-Based Study of 0.9 Million Norwegian Women

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In Norway, single-cohort vaccination with quadrivalent human papillomavirus (qHPV) vaccine targeting 12-year-old girls took place in 2009–2016. In 2020, the oldest vaccinated cohort was 23 years old and had approached the age where risk of being diagnosed with cervical intraepithelial neoplasia grade 2 or worse (CIN2+) increases rapidly. The aim of this cohort study was to assess direct qHPV vaccine effectiveness (VE) against CIN2+ among Norwegian women aged 16–30 years in 2007–2020. By using population-based health registries and individual-level data on vaccination status and potential subsequent CIN2+ incidence, we found 82% qHPV VE among women vaccinated before age 17 years.

Keywords. cervical high-grade lesions; human papillomavirus (HPV); qHPV vaccine; vaccine effectiveness; population-based registry data.

In 2006, the US Food and Drug Administration and European Medicines Agency approved the use of Gardasil, a quadrivalent human papillomavirus (qHPV) vaccine that protects against infection and certain diseases caused by human papillomavirus (HPV) types 6, 11, 16, and 18. It has shown high efficacy in clinical trials [1] as well as high impact and effect in population-based studies on cervical precursors and genital warts [2], and even already on cervical cancer [3, 4].

In Norway, the decrease in anogenital warts has been attributed to the national school-based HPV vaccination program with the qHPV vaccine, initiated for 12-year-old girls (the 1997 birth cohort) in 2009 [5]. From 2016 to 2019, the bivalent vaccine targeting HPV16 and HPV18 was used in a catch-up program for women 19–26 years of age (1990–1996 cohorts), and from 2017 the bivalent vaccine replaced the qHPV vaccine in the school-based program (Figure 1). While the vaccine coverage in the target group has increased from 75% to over 90%, the lack of catch-up vaccination until 2016 implies that girls born before 1997 were vaccinated at a suboptimal age (after the age of 19 years), which may have reduced the impact of HPV vaccination in Norway.

In Norway, the incidence of cervical intraepithelial neoplasia grade 2 or worse (CIN2+) starts to increase before age 20 years and peaks around age 26 years [6]. The single-cohort vaccination scheme for girls from 2009 to 2016 has led to the first vaccinated cohort reaching the age of 23 years in 2020, thus approaching screening age, which is from 25 years of age in Norway [7]. In this report, we are leveraging this opportunity to conduct the initial assessment of the effectiveness of the qHPV vaccine in Norway. The present observational cohort study aims to estimate the effectiveness of the qHPV vaccine against CIN2+ in Norway using population-based registry data from 2007 to 2020.

METHODS

Study Population

We used the National Register to obtain individual-level information on all 1 069 607 Norwegian female residents aged 16 to 30 years alive from 1 January 2007 through 31 December 2020. Women's unique personal identification number (PIN), date of birth, immigration, emigration, and death (if applicable) were obtained. A PIN is assigned to every individual at birth (or at immigration) and does not change through the resident's lifetime, ensuring accurate linking between the data sources.

Exposure

To obtain data on the type and date of HPV vaccines administered, we used the Norwegian Immunization Registry and the Norwegian Prescribed Drug Registry, which together contain complete information on all the vaccines administrated. For each woman in our cohort, we obtained the date and type of administrated HPV vaccine by using Anatomical Therapeutic Chemical codes (ATC) J07BM01 (qHPV vaccine), ATC J07BM02 (bivalent HPV vaccine), and ATC J07BM03 (nonavalent HPV vaccine).

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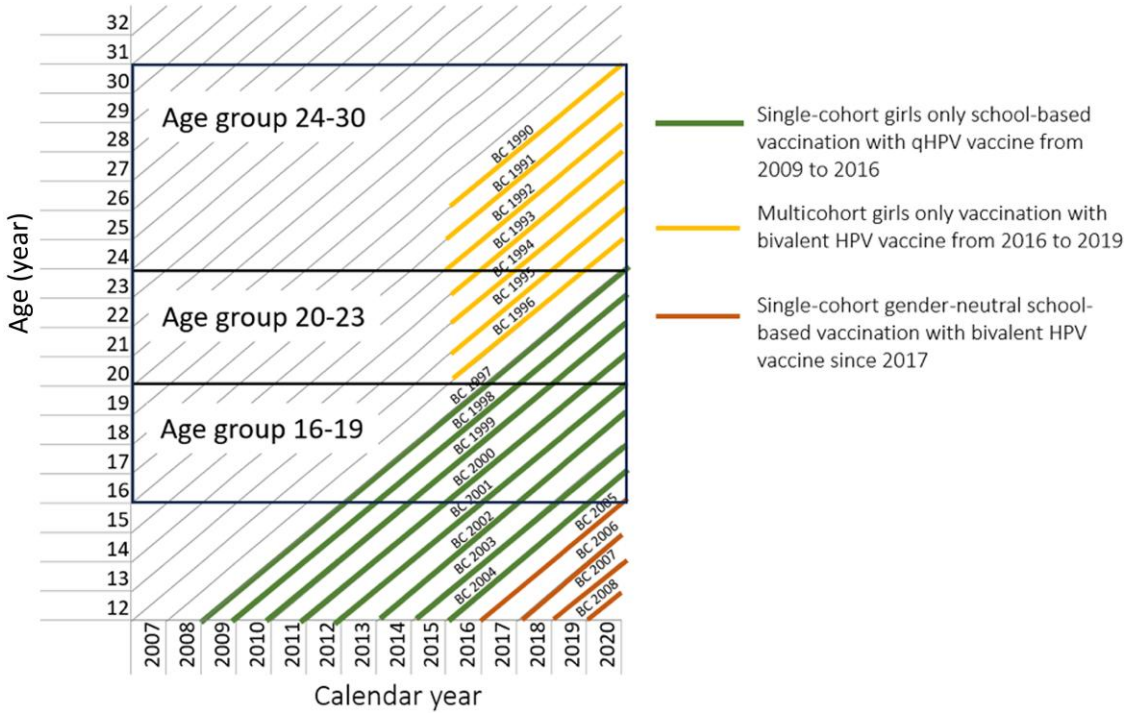


Figure 1. Included ages, age groups, and national HPV vaccination program setting in Norway during the study period from 2007 to 2020. Abbreviations: BC, birth cohort; qHPV, quadrivalent human papillomavirus.

Outcome

For CIN2+ incidence data, we linked all study participant with the Cancer Registry of Norway, which incorporates information from the Norwegian Cervical Cancer Screening Program (NCCSP). The NCCSP data covers screening activities, follow-up, and outcomes for women aged 16 years and older. Both databases have high data completeness, quality, and validity [8, 9]. For each woman, we identified the date of the first occurrence of a histologically confirmed diagnosis of CIN2+, including CIN2, CIN3, carcinoma in situ, adenocarcinoma in situ, and invasive cervical cancer.

Assessing Person-Years

We excluded women who immigrated to Norway after the age of 9, as we would not know if these women were vaccinated prior to immigration. We also excluded women who initiated vaccination of qHPV prior to 2007, or who had a diagnosis of CIN2+ prior to individual follow-up. We counted person-years for every individual from their 16th birthday or from 1 January 2007 until CIN2+ diagnosis, their 30th birthday, emigration, death, or 31 December 2020, whichever came first. Depending on the date of vaccination, a woman could enter the follow-up either unvaccinated or vaccinated. Women were counted as unvaccinated until the date of first dose of qHPV vaccine administration and thereafter as vaccinated. Women who received an HPV vaccine other than the qHPV vaccine contributed person-years as unvaccinated until the vaccination date and were censored after that.

Statistics

CIN2+ overall incidence rates (IR) were calculated for any vaccination age and a separate analysis was performed for women who were vaccinated before the age of 17 years, which corresponds to the median age of sexual debut among Norwegian girls [9]. In addition, stratified IRs were calculated according to the CIN2+ attained age in age groups 16–19, 20–23, and 24–30 years. IRs were calculated by dividing the number of new cases of CIN2+ by the corresponding person-years accumulated in the study period according to vaccination status as defined above. IRs and corresponding 95% confidence intervals (CI) were calculated by Poisson regression with the natural log of person-time as an offset. The model included age as a continuous covariate to account for the age-dependent probability of CIN2+. Calendar year was also included as a continuous covariate to adjust for potential incidence trends over time. We calculated incidence rate ratios (IRR) between the groups. Vaccine effectiveness (VE) was calculated $(1 - \text{IRR}) \times 100\%$. Sensitivity analyses were performed assuming (1) the vaccine to be effective after the third dose (and not after the first dose), (2) adding a lag time of 6 months from vaccination to vaccine effect, and (3) using CIN3+ as outcome. The statistical software Stata 17/18 was used for all analyses.

Table 1. Incidence of CIN2+ in the Total Population of Norwegian Women Aged 16–30 Years During 2007–2020, by Vaccination Status (at Least 1 Dose of Gardasil), CIN2+ Attained Age, and Age at Vaccination

Attained Age, y/Age at First Dose of Gardasil, y	CIN2+[a]					
	No. of Cases	PY	IR (95% CI)	Adjusted IR[b] (95% CI)	Adjusted IRR (95% CI)	Adjusted VE (95% CI)
Unvaccinated						
Overall	18 098	4 922 140.1	367.7 (362.3–373.0)	183.7 (179.3–188.2)	Reference	Reference
Attained age 16–19	158	1 192 563.1	13.2 (11.2–15.3)	12.9 (10.5–15.3)	Reference	Reference
Attained age 20–23	2191	1 467 951.1	149.3 (143.0–155.5)	142.8 (136.7–148.8)	Reference	Reference
Attained age 24–30	15 749	2 249 211.0	700.2 (689.3–711.1)	615.9 (605.8–626.0)	Reference	Reference
Vaccinated[c]						
Overall	626	876 473.2	71.4 (65.8–77.0)	72.1 (65.8–78.4)	0.39 (.36–.43)	61 (57–64)
Attained age 16–19	21	618 420.1	3.4 (1.9–4.8)	2.8 (1.4–4.2)	0.22 (.12–.38)	78 (62–88)
Attained age 20–23	190	219 562.5	86.5 (74.2–98.8)	53.8 (45.1–62.4)	0.38 (.32–.45)	62 (55–68)
Attained age 24–30	415	38 493.6	1078.1 (974.4–1181.8)	586.9 (523.6–650.2)	0.95 (.86–1.06)	5 (−6 to 14)
Vaccinated,[c] stratified by age at first dose vaccinated < 17.0 y						
Overall	225	827 443.4	27.2 (23.6–30.7)	35.0 (30.3–39.7)	0.18 (.16–.21)	82 (79–84)
Attained age 16–19	17	613 544.0	2.8 (1.5–4.1)	2.3 (1.1–3.5)	0.18 (.10–.31)	82 (69–90)
Attained age 20–23	151	203 549.4	74.2 (62.4–86.0)	45.9 (37.8–53.9)	0.32 (.27–.39)	68 (61–73)
Attained age 24–30	57	10 352.5	550.6 (407.7–693.5)	287.4 (212.5–362.3)	0.47 (.36–.61)	53 (39–64)

Abbreviations: CI, confidence interval; CIN, cervical intraepithelial neoplasia; IR, incidence rate reported per 100 000 PY; IRR, incidence rate ratio; PY, person-years; VE, vaccine effectiveness.

[a]CIN2+ indicates CIN2, CIN3, adenocarcinoma in situ, or cervical cancer; if a subject had multiple diagnoses during the study period, person-years were calculated according to the first occurrence of histologically confirmed CIN2+ case.

[b]Adjusted for calendar year. Overall estimates also adjusted for CIN2+ attained age.

[c]Vaccinated indicates the person-years after receipt of the first dose of qHPV vaccine until the end of individual follow-up.

RESULTS

In total, 868 403 women with 5 798 051 person-years contributed to the analyses. From 2007 to 2020, 18 098 CIN2+ events occurred among unvaccinated (4 922 140 person-years) and 626 CIN2+ events among vaccinated (876 473 person-years) individuals aged 16–30 years in Norway (Table 1). The IRs were lower for vaccinated women, resulting in an overall VE of 61% (95% CI, 57%–64%). VE was highest among females in the youngest age group (16–19 years) at 78% (95% CI, 62%–88%). For age groups 20–23 and 24–30 years, the VE was 62% (95% CI, 55%–68%) and 5% (95% CI, −6% to 14%), respectively.

When we limited analyses to those women who received at least 1 dose of qHPV vaccine before the age of 17 years, the overall VE against CIN2+ was estimated to be 82% (95% CI, 79%–84%). Among those women, qHPV VE at the age groups of 16–19, 20–23, and 24–30 years was 82% (95% CI, 69%–90%), 68% (95% CI, 61%–73%), and 53% (95% CI, 39%–64%), respectively.

The sensitivity analyses showed slightly increased VEs both for (1) assuming the vaccine not to be efficient before the third dose, (2) adding a lag time of 6 months before the vaccine was effective, and (3) using CIN3+ as outcome (Supplementary Material).

DISCUSSION

Our study contributes to a growing body of evidence that the qHPV vaccine is highly effective in preventing histologically confirmed CIN2+ lesions. The high VE of 82% was observed among those who received the vaccine before the age of 17 years, the estimated median age of sexual debut in Norway [10].

Our results are in line with other population-level studies from Nordic countries, utilizing individual-level data from national health registries to estimate the VE. Studies from Sweden and Denmark report 75% and 57%, respectively, overall VE against CIN2+ for those who initiated vaccination before the age of 17, whereas our estimate is 82% [11, 12]. The higher VE in Norway compared to Denmark in particular, but also Sweden, may be due to the multicohort vaccination in Denmark and Sweden, as a larger proportion of the unvaccinated girls in the Swedish and the Danish studies are from vaccinated cohorts who benefit from herd protection. In the Norwegian setting, almost all unvaccinated females are from the cohorts that are not impacted by intracohort herd protection, resulting in larger IR difference between the cohorts and hence higher VE. This effect is most likely attenuated by Norway's more stable and higher vaccine coverage in the younger age groups targeted through school-based programs (estimated to be 92%, compared to 79% in Denmark and 86% in Sweden in 2019 [13]), resulting in very few unvaccinated females coming from childhood vaccinated cohorts (and who benefit from intracohort herd protection) in the Norwegian study compared to the Swedish and Danish studies.

Also contributing to the high VE in our study is the fact that women in our cohort are still young and CIN cases that are occurring in this age are mostly attributed to HPV16 and 18 [14]. With increasing age, the attribution of other HPV types contributing to cervical disease increases, which would be expected to result in lower observed VE over time in older age groups [14].

We note that for those vaccinated before age of 17 years, the VE for overall (any age) is very similar to the VE for the youngest age group (16–19 years). This is because most women vaccinated before age 17 years in this study were from the youngest birth cohorts (from 1997 and onwards), who did not reach the oldest age group. We also need to note that modelling for overall and by age groups was done separately, so the adjustments factor (calendar time) may be different between the analyses.

Lack of VE against CIN2+ for the age group 24–30 years can be related to the fact that none of the birth cohorts eligible for school-based vaccination reached 24 years of age in this study. This means that this age group was merely opportunistically vaccinated and tended to have higher vaccination age than the other age groups. This result is in line with previous studies showing little or no effect of opportunistic vaccination, especially for higher vaccination ages [15]. It is also important to consider that among women vaccinated at older ages, it may take longer time to observe high VE, as most disease that occurs soon after vaccination would be attributable to infections that were acquired at younger age and ongoing at the time of vaccination. To account for ongoing infections, for which the vaccine may not be efficient, we included a buffer period of 6 months from vaccination to assumed vaccine effect. We also performed analyses where we assumed the vaccine not to be effective before the third dose, thus further extending the buffer period by about 6 months from the first dose. The results showed an increase in the overall VE from 5% to 16% for the oldest age group (Supplementary Material).

Our study benefits from a large nationwide study cohort, using data from over 13 years and including 15 birth cohorts. We used high-quality registries and collected individual-level data, which allowed us to estimate the effectiveness of the qHPV vaccine on CIN2+. However, the epidemiology of high-grade cervical lesions is not always straightforward. It is impacted by factors such as screening coverage, sensitivity of the screening technology, and general background risks such as HPV prevalence in the population.

Future planned analyses with longer follow-up, including women from vaccinated cohorts invited to the regular screening program in Norway, will provide more precise estimates of VE.

In conclusion, school-based qHPV vaccination in Norway, despite being a single-cohort and single-sex vaccination scheme from 2009 to 2016, has shown remarkable VE.

Supplementary Data

Supplementary materials are available at *The Journal of Infectious Diseases* online (http://jid.oxfordjournals.org/). Supplementary materials consist of data provided by the author that are published to benefit the reader. The posted materials are not copyedited. The contents of all supplementary data are the sole responsibility of the authors. Questions or messages regarding errors should be addressed to the author.

Notes

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Potential conflicts of interest. J. E. T. is an employee of Merck Sharp & Dohme, LLC, a subsidiary of Merck & Co, Inc, Rahway, NJ. The affiliated institute of M. O., E. J. L., M. G., M. N., and S. N. has received research grants from MSD (Norge) AS.

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